

0770/3/2024
P.M.S. A/L

**SOUTH WEST REGIONAL MOCK EXAMINATION
GENERAL EDUCATION**

THE TEACHERS' RESOURCE UNIT (TRU)

IN COLLABORATION WITH

THE REGIONAL INSPECTORATE OF PEDAGOGY FOR SCIENCE EDUCATION

AND

THE SOUTH-WEST ASSOCIATION OF MATHEMATICS TEACHERS

(SWAMT)

MONDAY AFTERNOON, 25/03/2024

ADVANCED LEVEL

Subject Title	Pure Mathematics With Statistics
Paper Number	Paper 3
Subject Code Number	0770

Three Hours

INSTRUCTIONS TO CANDIDATES:

Answer ALL questions.

For your guidance, the approximate mark allocation for parts of each question is indicated in brackets.

You are reminded of the necessity for good English and orderly presentation in your answers.

Mathematical formulae and tables published by the CGCE Board and noiseless, non-programmable calculators are allowed.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage

1. (i) Events A and B are such that $P(A) = \frac{1}{3}$, $P(B) = \frac{2}{5}$ and $P(A/B) = \frac{1}{4}$. Find,
- (a) $P(A \cap B)$ (2 marks)
 - (b) $P(A \cup B)$ (2 marks)
 - (c) $P(B/A')$ (3 marks)
- (ii) In a certain school 30% of the students are boys and 70% are girls. 5% of the boys and 10% of the girls use reading glasses. A student is selected at random from the school. Using a probability tree diagram or otherwise, find
- (a) The probability that the student is uses reading glasses. (3 marks)
 - (b) The probability that the student is a boy given he uses reading glasses. (3 marks)

2. The masses, to the nearest kilogram, of 60 students in a certain high school were measured and recorded as follows :

Mass, x	26 - 30	31 - 35	36 - 40	41 - 45	46 - 50	51 - 55	56 - 60
No of students	5	10	18	12	8	4	3

- (a) Calculate, to one decimal place, the mean and standard deviation of the masses. (7 marks)
- (b) Draw a cumulative frequency curve for the distribution. (2 marks)
- (c) From the curve, find the quartile deviation of the masses. (4 marks)

3. A discrete random variable X has the following distribution :

x	1	2	3	4	5
$P(X = x)$	$\frac{1}{5}$	a	$\frac{3}{10}$	b	$\frac{3}{20}$

Given that $11P(X \leq 2) = 9P(X > 2)$,

- (a) Find the values of the probabilities a and b. (6 marks)
- Using these values,
- (b) show that $E(X) = \frac{11}{4}$, (3 marks)
 - (c) find the variance of X. (4 marks)

4. A continuous random variable, X, has probability density function, f, defined by

$$f(x) = \begin{cases} \frac{1}{k} (k - x), & 1 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Find,

- (a) the value of the constant k, (4 marks)
- (b) the mean and variance of this distribution, using the value of k above, (6 marks)
- (c) $P(1.5 \leq X \leq 2.6)$ (3 marks)

5. (i) In a certain school 25% of form five students passed the Regional Mock Examination. 5 students are randomly selected from the class, find the probability that
- (a) exactly two passed the Regional Mock (2 marks)
 - (b) at least one passed the Regional Mock (3 marks)

- (ii) It is observed that 0.2% of maize seeds planted are eaten up by birds. Using the Poisson Distribution, find, correct to four decimal places, the probability that out of 2,000 seeds planted,
- (a) none will be eaten up, (3 marks)
 - (b) exactly three will be eaten up, (2 marks)
 - (c) more than three will be eaten up. (3 marks)

-
6. (i) The following samples are drawn from a given population.

Sample 1: $n_1 = 1$, $\Sigma(x_1) = 422$, $\Sigma(x_1^2) = 22412$

Sample 2: $n_2 = 10$, $\Sigma(x_2) = 523$, $\Sigma(x_2^2) = 27532$

Using this data, find the unbiased estimates of the population mean and variance (6 marks)

- (ii) A light bulb manufacturing industry claims that their bulbs have a life span of 270 days, with a standard deviation of 12 days. The industry increases quality, with the intension of increasing the life span of their bulbs. A random sample of 81 bulbs tested gave a mean life span of 274 days. By stating the null and alternative hypothesis, determine at the 5% level of significance, whether this sample provides sufficient evidence for an increase in quality and conclude accordingly. (7 marks)

-
7. In a certain school, the masses of male students, X , are approximately normally distributed, with mean 60 kg and standard deviation 5 kg. If 36 male students are selected at random, find the probability that their mean mass

- (a) is 61 kg or more (3 marks)
- (b) lies between 59 kg and 63 kg. (4 marks)

A Sample of 49 female students from the same school is selected and the masses of the female students, Y , are also approximately normally distributed, with mean 65 kg and standard deviation 4 kg.

- (c) Find the distribution of $(\bar{X} + \bar{Y})$ (2 marks)
 - (d) Find $P[(\bar{X} + \bar{Y}) > 124]$ (4 marks)
-

8. The marks obtained by eight students in Biology and Chemistry are shown below:

Biology marks, x	31	32	37	42	43	49	52	54
Chemistry marks, y	20	18	25	30	29	33	36	35

Calculate, correct to three decimal places,

- (a) Pearson's product-moment correlation coefficient for these data, (5 marks)
- (b) the minimum sum of squares of residuals for x on y and for y on x , (4 marks)
- (c) the least squares regression line of Chemistry marks on Biology marks. (2 marks)
- (d) the chemistry marks for a candidate who scored 46 in Biology. (2 marks)

You may assume that $\sum(x) = 340$, $\sum(y) = 226$, $\sum(xy) = 10,007$, $\sum(x^2) = 14,988$ and $\sum(y^2) = 6,700$.

END